Fertility and Housing Tenure Choice:
Model Writeup

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1 Model

The objective is to model a lifecycle economy with endogenous fertility, housing tenure choice, and spatial sorting. Households choose completed fertility, decide where to live, rent or own housing, and save over the lifecycle. Children enter preferences and resources directly through consumption and housing requirements. Housing is local, supplied upward sloping in each location, and costly to access through down-payment and moving frictions. The central mechanism is therefore simple: children require space; space is more expensive in central locations; and tenure constraints affect the timing and location of family formation.

The environment brings together three standard ingredients. The fertility block follows quantitative lifecycle models in which children are chosen by households and affect both current resources and continuation values (Sommer, 2016; Doepke and Kindermann, 2019). The tenure block follows housing-macro models with collateral constraints and discrete owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The spatial block is Rosen–Roback: households sort across locations with different amenities and prices, and local housing supply determines how demand pressure maps into prices (Rosen, 1979; Roback, 1982). The housing structure is very heavily inspired by Greaney et al. (2025).

1.1 Environment

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R [1], and die at the end of age J. At a point in time the city is described by a level distribution g over household states. Its adult mass is

$$S = \int dg(x).$$

This mass is an equilibrium object. When useful, I write $g = S\mu$ with $\int d\mu = 1$.[2]

[1] I might consider simplifying the lifecycle structure to just some stages, to improve computational efficiency. The next step is, indeed, clearly determining what are the empirical targets for the model, and so whether I want to match very well lifecycle profiles or not.

[2] More on this later. I understand this is the main 'technical' issue that emerged in the seminar. It is not fully solved. At the end of this document there is a discussion of the things. I hope to get feedback on this.

There are two residential locations, $\mathcal{I} = \{P, C\}$, where P denotes the periphery and C denotes the center. A unit of housing in location i has purchase price p_i and rental price $r_i = (q + \delta)p_i$, where q is the net interest rate and δ is depreciation or maintenance. Location i also has amenity $\mathcal{E}_i$, housing supply scale H_{0i}, and housing supply elasticity η_i. The amenity $\mathcal{E}_i$ enters the location choice value, H_{0i} shifts the level of local housing supply, and η_i governs the slope of that supply curve. The center is the high-amenity, high-price location in the benchmark.

The individual state at the beginning of a period is $x = (b, d, i, a, n, s)$. Here b is liquid wealth, d is tenure, i is current location, a is age, n is completed fertility, and s is the child-age state. Tenure belongs to $\mathcal{D} \equiv \{R, O_1, \ldots, O_{N_H}\}$. R denotes renting. O_k denotes ownership of a house with physical size H_k, where $H_1 < \cdots < H_{N_H}$. Owners choose from this discrete ladder [3]. Renters choose rental space continuously up to a cap $\bar{h}^R$, which is the largest rental unit available in the model.

1.2 Preferences

Let $m(n, s)$ be the number of dependent children living at home [4]. Flow utility is Stone–Geary over nondurable consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha} \left(\mathbf{h} - \bar{h}(n, s)\right)^{1-\alpha}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m(n, s), \tag{1}$$

where $\alpha \in (0, 1)$ is the expenditure-share parameter inside the consumption-housing aggregator, $\sigma > 1$ is curvature, and ψ_{child} is the flow value of each dependent child.[5] The Stone–Geary terms $\bar{c}(n, s)$ and $\bar{h}(n, s)$ are minimum consumption and housing needs:

$$\begin{aligned}
\bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\
\bar{h}(n, s) &= \bar{h}_0 + \mathbb{I}\{m(n, s) \geq 1\}\bar{h}_{\text{jump}} + \bar{h}_n m(n, s).
\end{aligned}$$

Here $\bar{c}_0$ and $\bar{h}_0$ are baseline needs for a childless household. The parameter $\bar{c}_n$ is the per-child consumption requirement. The parameter $\bar{h}_{\text{jump}}$ is the fixed housing requirement associated with having any dependent child, while $\bar{h}_n$ is the additional housing requirement per dependent child. The first child therefore raises required housing discretely; additional children raise required housing linearly. This is the main channel in the paper: children are costly partly because they require housing space, and the price of that space depends on location.

Housing services differ by tenure. A renter choosing rental space h^R receives $\mathbf{h} = h^R$. An owner of house size H_k receives $\mathbf{h} = \chi H_k$, with $\chi > 1$. The parameter χ captures the service premium of owner-occupied housing. In market clearing, however, owners demand the physical quantity H_k, not the service quantity χH_k.

At death, households receive warm-glow bequest utility. Let $\psi \in [0, 1)$ denote the proportional

[3] This makes housing more lumpy and makes the model computationally lighter, but might be amended. Some further thinking is required on the housing market structure.

[4] Some notation improvement is probably required, but in any case, the presence of both m and n allows us to distinguish children ever had with children currently at home. which affects the utility.

[5] The position of the ψ_{child} term determines how the child benefit relates to income, so it requires some further thought.

transaction cost paid when owner-occupied housing is liquidated. Bequeathable wealth is

$$\widetilde{b} = \begin{cases} b, & d = R, \\ b + (1-\psi)p_i H_k, & d = O_k. \end{cases}$$

The bequest term is

$$\mathcal{B}(\widetilde{b}, n) = \theta_0(1+\theta_n n)\frac{(\theta_1 + \max\{\widetilde{b}, 0\})^{1-\sigma}}{1-\sigma}. \tag{2}$$

The parameter θ_0 governs the overall strength of the bequest motive. The parameter θ_n scales the bequest motive with completed fertility, so completed fertility affects bequests even after children leave the household. The shift θ_1 controls curvature near zero wealth. This allows parenthood to matter for late-life saving and ownership, not only for years with dependent children.

1.3 Firms, Income, and Assets

Competitive firms produce the consumption good with aggregate efficiency labor, $Y = AL$. The consumption good is the numeraire. Competition implies $w = A$, where w is the common wage per efficiency unit. Wages do not vary by location, so spatial sorting is driven by amenities, housing costs, tenure frictions, and moving wedges, not by local wage differences.

Workers supply labor inelastically. A worker of age a supplies e_a efficiency units, so labor income is

$$y_a = (1-\tau_{\text{pay}})we_a, \qquad a \le J_R,$$

where e_a is the deterministic age-efficiency profile and τ_{pay} is the payroll tax rate. Retired households receive

$$y_a = p^{\text{ret}}, \qquad a > J_R.$$

The pension benefit is determined in equilibrium by the pay-as-you-go government budget.

Given the level distribution g, aggregate efficiency labor supply is

$$L^s(g) = \int \mathbb{I}\{a \le J_R\} e_a \, dg(x).$$

In equilibrium firms hire this labor supply, so $L = L^s(g)$ and output is $Y = AL$.

Liquid wealth earns gross return $\mathcal{R} = 1 + q$. A renter cannot borrow:

$$b' \ge 0.$$

An owner can borrow against the house:

$$b' \ge -\phi p_i H_k. \tag{3}$$

Here ϕ is the financed share of house value. The required down payment is therefore $(1-\phi)p_i H_k$.

1.4 Housing and Budgets

Given $r_i = (q + \delta)p_i$, a renter chooses c, h^R, and b' subject to

$$c + r_i h^R + b' = \mathcal{R}b + y_a, \qquad 0 \leq h^R \leq \bar{h}^R, \qquad b' \geq 0.$$

An owner of house H_k chooses c and b' subject to

$$c + \delta p_i H_k + b' = \mathcal{R}b + y_a, \qquad b' \geq -\phi p_i H_k.$$

Mortgage debt is negative liquid wealth, so mortgage interest is already embedded in $\mathcal{R}b$. Owners pay maintenance directly.

Housing adjustment is costly. The same transaction-cost parameter ψ used in bequeathable wealth applies when a household sells a house. Selling a house in location i yields net proceeds

$$(1 - \psi)p_i H_k.$$

A moving owner sells the origin house before choosing tenure in the destination. A stayer can keep the current house, sell and rent, or move to another owner rung if the budget and collateral constraints allow it.

1.5 Fertility and Child Aging

During the fertile window

$$a \in \{A_f^{\text{start}}, \ldots, A_f^{\text{end}}\},$$

a childless household chooses $n' \in \mathcal{N} \equiv \{0, 1, \ldots, \bar{n}\}$. The bounds A_f^{start} and A_f^{end} define the ages at which fertility can occur, and $\bar{n}$ is the maximum completed fertility choice. The choice $n' = 0$ means no birth in the current period. A positive choice $n' > 0$ fixes completed fertility forever. A household that reaches the end of the fertile window without a positive fertility choice remains childless. Children are born together and age together through K child stages. This is not a sequential parity-progression model. A model object that looks like a second-child housing response should therefore be interpreted as an additional-child housing-demand moment, not as a second-birth hazard.

Let $s = 0$ denote no children and $s = 1, \ldots, K$ denote dependent child stages. After the last dependent stage, children mature and leave the household. Child aging follows

$$\Pi^c(s'|s, n).$$

The matrix Π^c is the transition matrix over child-age states, conditional on completed fertility. In the benchmark implementation, dependent childhood lasts about 18 years. While children are at home, $m(n, s) = n$. After maturity, $m(n, s) = 0$, so the consumption and housing floors return to their baseline levels, while completed fertility remains payoff-relevant through the bequest motive. The same aging matrix determines the flow of city-born children who become young adults:

$$B(g) = \int n \Pr_{\Pi^c}\left(s' \notin \{0, 1, \ldots, K\} \mid s, n\right) dg(x). \tag{4}$$

Mature city-born children are measured in entrant-household equivalents and are counted as potential entrants; the outside option determines whether they enter the modeled city.

1.6 Household Problem

Within the period, the order of decisions is fertility, location, tenure and house size, and then consumption, rental size, and savings. The computation solves the problem backward in the reverse order. The discount factor is β. Let the terminal value be

$$V_{J+1}(b, d, i, n, s) = \mathcal{B}(\widetilde{b}, n),$$

and define the continuation value after child aging by

$$\bar{V}_{a+1}(b', d', i', n, s) = \sum_{s'} \Pi^c(s'|s, n) V_{a+1}(b', d', i', n, s').$$

Conditional on being a renter in location i, the household value is

$$\begin{aligned} V_a^R(b, i, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R; n, s) + \beta \bar{V}_{a+1}(b', R, i, n, s) \\ \text{s.t.} \quad & c + r_i h^R + b' = \mathcal{R} b + y_a, \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{5}$$

Conditional on owning house H_k in location i, the value is

$$\begin{aligned} V_a^{O_k}(b, i, n, s) = \max_{c, b'} \quad & u(c, \chi H_k; n, s) + \beta \bar{V}_{a+1}(b', O_k, i, n, s) \\ \text{s.t.} \quad & c + \delta p_i H_k + b' = \mathcal{R} b + y_a, \\ & b' \ge -\phi p_i H_k. \end{aligned} \tag{6}$$

The housing-adjustment problem follows the same accounting as in Greaney et al. (2025). Let $H(d) = 0$ if $d = R$ and $H(d) = H_k$ if $d = O_k$. If a household in state (b, d, i) chooses residence j, its liquid wealth and owner-occupied house after the residence choice are

$$\widetilde{b}^R(b, d, i; j) = b + \mathbb{I}\{j \neq i\}(1 - \psi) p_i H(d), \qquad \widetilde{H}^R(d, i; j) = \mathbb{I}\{j = i\} H(d).$$

A moving homeowner sells the origin house and pays the transaction cost. A household that remains in the same location carries its owner-occupied house into the housing-adjustment stage. If it then chooses tenure d' in location j, liquid wealth becomes

$$\widetilde{b}^H(b, d, i; j, d') = \widetilde{b}^R(b, d, i; j) + \mathbb{I}\{H(d') \neq \widetilde{H}^R(d, i; j)\} \left[(1 - \psi) p_j \widetilde{H}^R(d, i; j) - p_j H(d')\right]. \tag{7}$$

The feasible tenure set in destination j is

$$\mathcal{D}(b, d, i; j) = \left\{ d' \in \mathcal{D} : \widetilde{b}^H(b, d, i; j, d') \ge \underline{b}_j(d') \right\},$$

where $\underline{b}_j(R) = 0$ and $\underline{b}_j(O_k) = -\phi p_j H_k$. The tenure value in destination j is

$$V_a^T(b, d, i; j, n, s) = \max_{d' \in \mathcal{D}(b, d, i; j)} V_a^{d'} \left(\widetilde{b}^H(b, d, i; j, d'), j, n, s \right). \tag{8}$$

Here $V_a^{d'} = V_a^R$ if $d' = R$ and $V_a^{d'} = V_a^{O_k}$ if $d' = O_k$.

Location choice has additive Type-I extreme-value shocks with scale κ_ℓ. The parameter κ_ℓ governs the amount of residual dispersion in location choices; as κ_ℓ goes to zero, location choice converges to a deterministic maximum. A household currently in location i evaluates destination j using

$$\widetilde{V}_a^L(j|x) = V_a^T(b, d, i; j, n, s) + \mathcal{E}_j - \zeta_{ij},$$

where ζ_{ij} is the bilateral moving wedge, with ζ_{ii} normalized to zero. The destination probabilities are

$$\pi_a^L(j|x) = \frac{\exp\left(\widetilde{V}_a^L(j|x)/\kappa_\ell\right)}{\sum_{m\in\mathcal{I}} \exp\left(\widetilde{V}_a^L(m|x)/\kappa_\ell\right)}, \tag{9}$$

and the location value is

$$V_a^L(x) = \kappa_\ell \log \sum_{m\in\mathcal{I}} \exp\left(\widetilde{V}_a^L(m|x)/\kappa_\ell\right). \tag{10}$$

Fertility choice has additive Type-I extreme-value shocks with scale κ_f. The parameter κ_f governs residual dispersion in completed-fertility choices. This shock is part of the household fertility problem, not part of the population-entry closure below. If a household is childless and in the fertile window, it compares $n' \in \mathcal{N}$. Let $s(n') = 0$ if $n' = 0$ and $s(n') = 1$ if $n' > 0$. The value of choosing completed fertility n' is

$$x(n') = (b, d, i, a, n', s(n')), \qquad \widetilde{V}_a^F(n'|x) = V_a^L(x(n'))).$$

Fertility probabilities are

$$\pi_a^F(n'|x) = \frac{\exp\left(\widetilde{V}_a^F(n'|x)/\kappa_f\right)}{\sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m|x)/\kappa_f\right)}, \tag{11}$$

and the full household value is

$$V_a(x) = \kappa_f \log \sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m|x)/\kappa_f\right). \tag{12}$$

Outside the fertility window, or after fertility has already been chosen, this stage is degenerate and $V_a(x) = V_a^L(x)$.

1.7 Entry, Outside Option, and City Size

In a model with fertility, some non trivial problems emerge when thinking about closure of population size. Here I give one possible solution, and at the end of the document I discuss a couple of other avenues. Let g denote the level distribution of adult households, and let

$$S = \int dg.$$

I write this distribution as

$$g = S\mu, \qquad \int d\mu = 1,$$

where μ is the composition of the city. The normalization is only a computational convenience; market clearing uses g.

The closure works through a pool of potential young-adult entrants. Each period the city receives an exogenous flow $M \geq 0$ of outside-born potential entrants. It also receives the mature city-born flow $B(g)$ generated by the child-aging matrix. Thus the potential-entrant pool is

$$Q(g) = M + B(g). \tag{13}$$

Actual entry from this pool is determined by the outside option below.

A potential entrant enters the modeled city only if the city is sufficiently attractive relative to the outside economy. The value of entering location i is

$$W_i^E(p) = V_1(b^E, R, i, 0, 0; p),$$

where entrants begin as childless renters with liquid wealth b^E. This is an endogenous object: it is computed from the household problem at prices p. The outside value $\bar{W}^E$ is exogenous. Potential entrants draw idiosyncratic Type-I extreme-value tastes over the modeled locations and the outside economy, with scale parameter κ_E. The unconditional probability of entering location i is

$$\pi_i^E(p) = \frac{\exp\left(W_i^E(p)/\kappa_E\right)}{\exp\left(\bar{W}^E/\kappa_E\right) + \sum_{m\in\mathcal{I}} \exp\left(W_m^E(p)/\kappa_E\right)}. \tag{14}$$

The total city-entry probability and conditional entrant location shares are

$$q^E(p) = \sum_{i\in\mathcal{I}} \pi_i^E(p), \qquad \omega_i^E(p) = \frac{\pi_i^E(p)}{q^E(p)}.$$

Entry into location i is

$$E_i(g;p) = Q(g)\pi_i^E(p) = Q(g)q^E(p)\omega_i^E(p). \tag{15}$$

The remaining mass $Q(g)(1 - q^E(p))$ stays in the outside economy.

This is not a balanced-growth model. The outside economy supplies potential entrants and absorbs those who do not enter the modeled city. Stationarity pins down a level of city population, not a growth rate: at that level, actual young-adult entry equals the entry flow required to reproduce the stationary composition. For a candidate price vector p, household policies imply a stationary composition $\mu(p)$. Let

$$E_0(p) = \int \mathbb{I}\{a = 1\} d\mu(p)$$

be the young-adult entry flow required to reproduce one unit of that composition, and let $B_0(p) = B(\mu(p))$ be the mature city-born flow per unit of city scale. Since $g = S\mu$, stationarity requires

$$SE_0(p) = q^E(p)\left[M + SB_0(p)\right]. \tag{16}$$

If

$$E_0(p) > q^E(p)B_0(p), \tag{17}$$

then the stationary city scale is

$$S(p) = \frac{q^E(p)M}{E_0(p) - q^E(p)B_0(p)}. \tag{18}$$

The denominator is the net replacement gap. Quantitatively, it governs the strength of the scale channel: if the gap is small, small changes in fertility or entry values imply large changes in city size; if it is comfortably positive, the scale channel is more muted. The quantitative implementation reports this gap, or equivalently $q^E(p)B_0(p)/E_0(p)$, as a diagnostic of the closure. If condition (17) fails, the maintained stationary closure does not admit a finite city. Fertility matters for scale because births become mature city-born potential entrants through Π^c. At a given price vector, a larger mature city-born flow raises the stationary level $S(p)$ required by the entry accounting, unless offset by the outside-option entry rule. Fertility affects housing prices because this population level is solved inside market clearing. Given prices p, household policies determine $\mu(p)$, $\widehat{H}_i^D(p)$, and the mature city-born flow; the closure determines $S(p)$; and housing markets clear in levels:

$$S(p)\widehat{H}_i^D(p) = H_i^S(r_i).$$

Higher housing prices lower entry values and reduce $q^E(p)$. The outside option is the valve that keeps the equilibrium stationary while allowing fertility to affect aggregate housing demand.

The baseline level still needs units. One can choose M or $\bar{W}^E$ so that the baseline equilibrium has a convenient scale, for example $S = 1$. This is a normalization of units, not a fixed-population restriction. In counterfactuals, M and $\bar{W}^E$ are held fixed, so changes in fertility, prices, and household policies can change S.

1.8 Housing Supply and Market Clearing

Housing supply in location i is

$$H_i^S(r_i) = H_{0i} r_i^{\eta_i}. \tag{19}$$

The scale H_{0i} captures reduced-form land costs, regulation, and other supply-side wedges. The elasticity η_i governs the construction response. Thus the supply side has only a level parameter and an elasticity in each location.

Aggregate physical housing demand is

$$H_i^D(g) = \int \left[\mathbb{I}\{d = R, i(x) = i\} h^R(x) + \sum_{k=1}^{N_H} \mathbb{I}\{d = O_k, i(x) = i\} H_k \right] dg(x). \tag{20}$$

If $g = S\mu$, this is

$$H_i^D(g) = S\widehat{H}_i^D(\mu).$$

Normalizing the composition μ is harmless. Clearing housing markets with $\widehat{H}_i^D$ alone would instead impose a fixed-mass city. In the benchmark closure, prices solve

$$S(p)\widehat{H}_i^D(p) = H_{0i} r_i^{\eta_i}, \qquad i \in \mathcal{I}. \tag{21}$$

Fertility affects housing prices through two margins. It changes per-household housing demand, $\widehat{H}_i^D(p)$, because children require space and affect tenure and location choices. It also changes scale, $S(p)$, because higher fertility raises the flow of mature city-born children entering the potential-entrant pool. With upward-sloping housing supply, both margins feed into equilibrium prices.

2 Stationary Equilibrium

The equilibrium is stationary in prices, policies, and the level distribution of adult households. Stationarity is imposed on g, not only on the normalized composition μ. Entry, aging, fertility, location choice, tenure choice, savings, and death reproduce the same level distribution every period.

Definition 1 (Stationary equilibrium)**.** *Given primitives*

$$\Omega = \{q, \beta, \sigma, \alpha, \bar{c}_0, \bar{c}_n, \bar{h}_0, \bar{h}_{\text{jump}}, \bar{h}_n, \psi_{\text{child}}, \theta_0, \theta_n, \theta_1,$$
$$\chi, \phi, \psi, \delta, \tau_{\text{pay}}, \mathcal{E}_i, A, e_a, \zeta_{ii'}, H_{0i}, \eta_i, A_f^{\text{start}}, A_f^{\text{end}}, \bar{n}, \Pi^c, M, \bar{W}^E, \kappa_E\}_{i,i' \in \mathcal{I}},$$

a stationary equilibrium is a collection

$$\mathcal{E}^* = \big(\{p_i, r_i\}_{i\in\mathcal{I}}, w, L, Y, p^{\text{ret}}, \{V_a, V_a^R, V_a^T, V_a^L, \widetilde{V}_a^L, \widetilde{V}_a^F\}_a, \{V_a^{O_k}\}_{a,k}, \{c, h^R, b', d', i', n'\},$$
$$\{\pi^L, \pi^F, \pi^E, q^E, \omega^E\}, g, \{E_i, H_i^D, H_i^S\}_{i\in\mathcal{I}}\big)$$

such that:

1. ***Firms and labor market.*** *The common wage satisfies $w = A$. Aggregate efficiency labor supply is*

$$L^s(g) = \int \mathbb{I}\{a \le J_R\} e_a \, dg(x),$$

and the labor market clears:

$$L = L^s(g).$$

Output is $Y = AL$.

2. ***Household optimization.*** *Given prices, amenities, and moving wedges, value functions solve the household problem in (5)–(6), with tenure choice (8), location probabilities (9), fertility probabilities (11), and terminal bequest utility (2). The policies c, h^R, b', d', i', and n' are associated optimal policies.*

3. ***Entry and outside option.*** *The child-aging matrix implies a mature city-born flow $B(g)$. The potential entrant pool is $Q(g) = M + B(g)$. Entry probabilities satisfy (14), the city entry rate is $q^E(p) = \sum_i \pi_i^E(p)$, and entry masses satisfy*

$$E_i(g;p) = Q(g) q^E(p) \omega_i^E(p), \qquad i \in \mathcal{I}.$$

The outside mass is $Q(g)(1 - q^E(p))$.

4. ***Stationary distribution.*** *The level distribution g over (b, d, i, a, n, s) is invariant under the household policies, fertility and location probabilities, child aging, deterministic aging, death at age J, and the entry masses $\{E_i(g;p)\}_{i\in\mathcal{I}}$. If Γ denotes the one-period forward operator,*

$$g = \Gamma(g; p, \{E_i(g;p)\}_{i\in\mathcal{I}}).$$

Total adult population is

$$S = \int dg(x),$$

and is an equilibrium outcome.

5. ***Housing markets.*** *In every location,*

$$H_i^D(g) = H_i^S(r_i), \qquad r_i = (q+\delta)p_i.$$

6. ***Government budget.*** *Payroll-tax revenue finances pension benefits:*

$$\tau_{\text{pay}} wL = p^{\text{ret}} \sum_i \sum_{a>J_R} g_i(a).$$

Here $g_i(a)$ is the marginal mass of households of age a living in location i.

3 Population Closure: Alternatives

The closure above is the benchmark used in this writeup. It is useful to separate it from nearby alternatives, because the rest of the household and housing environment can be held fixed while the interpretation of counterfactuals changes.

Fixed adult mass. The simplest closure sets $\int dg = 1$ and keeps adult population fixed. This is common in lifecycle fertility models and in partial-equilibrium fertility exercises. Fertility then changes the composition of households, but not the mass of households in the city. Housing demand changes because parents want more space, sort differently, and choose tenure differently. It does not change because the city becomes larger. This is close in spirit to exercises that take the adult demographic envelope or prices as given, including Moreno-Maldonado and Santamaria (2024) and Karabarbounis and Nord (2025). The closure is clean, but it removes the scale channel that is central for aggregate local housing demand.

Reduced-form renewal valve. A second possibility is to ignore entry values and keep only demographic accounting:

$$SE_0(p) = M + \rho S B_0(p), \qquad S = \frac{M}{E_0(p) - \rho B_0(p)}.$$

The city receives an outside flow M and retains a fixed fraction ρ of mature city-born children. This delivers a finite stationary scale without introducing a transition path or balanced growth. Fertility affects aggregate housing demand through S. The limitation is that retention is mechanical: city attractiveness affects fertility and housing choices, but it does not directly affect the share of potential entrants who enter the city.

Outside-option entry. The benchmark in this note uses the outside-option version. It keeps the same stationary accounting logic, but makes city entry depend on the value of entering the city relative to an outside economy:

$$SE_0(p) = q^E(p)\left[M + SB_0(p)\right].$$

This is still a stationary model. It solves for a level S, not for a population growth rate. The difference from the reduced-form valve is that housing prices discipline city size through entry values. If fertility raises the flow of mature city-born children, the potential-entrant pool rises. If the larger city raises housing prices, entry values fall and the outside option absorbs more of the potential flow. This gives the model a scale channel without turning the paper into a theory of city growth.

Full demographic city growth. The most structural alternative is a sequence model in which births today become adult households in future periods, the age distribution evolves over calendar time, and housing markets clear along the transition. This is the natural object for a paper about city growth and fertility dynamics, and it is the direction taken by Coeurdacier et al. (2023). It is a different equilibrium object. One must take a stand on the path of age masses, migration flows, survival, and housing supply over time. The object here is instead a stationary city in which fertility, housing demand, tenure, and sorting are jointly determined.

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